HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Dupuytren and rheumatoid hand

Part A: Dupuytren disease

1. Epidemiology and associations

- **Benign fibroproliferative disorder of the palmar (and digital) fascia** leading to **nodules, cords and flexion contractures**.
- **Northern European (Celtic/Viking) ancestry**, **men (about 6-10:1)**, **older age (over 50)**, **autosomal dominant with variable penetrance**; **less common in Asians and Africans** (but seen in India).
- **Associations: alcohol (liver disease)**, **smoking**, **diabetes** (**milder: ring and middle fingers, palmar nodules**), **epilepsy (anticonvulsants: phenytoin, phenobarbital)**, **HIV**, **manual labour/vibration (debated)**, **hyperlipidaemia**, **previous hand trauma**.
- **Dupuytren diathesis (Hueston) (aggressive disease, high recurrence):**
  - **Early onset (under 50)**,
  - **Family history (positive)**,
  - **Bilateral disease**,
  - **Ectopic lesions: Garrod's knuckle pads (dorsal PIPJ)**, **Ledderhose disease (plantar fibromatosis)**, **Peyronie disease (penile fibromatosis)**,
  - **Male sex, Northern European ethnicity, radial-sided disease**.

2. Pathology

- **Myofibroblast** (**alpha-smooth muscle actin**: contractile) is the key cell.
- **Increased type III collagen (relative to type I)** (normal fascia: mainly type I).
- **Growth factors: TGF-beta (key), PDGF, basic FGF**; **Wnt signalling** genetic loci; **TNF** (anti-TNF trials for early nodules).
- **Luck stages:**
  - **Proliferative** (myofibroblasts; nodules; highly cellular),
  - **Involutional** (myofibroblasts align along tension lines; cords form),
  - **Residual** (acellular, tendon-like cords; fibrocytes, type I collagen).

3. Normal fascia becomes diseased cords

Normal structure	Diseased cord	Effect
Pretendinous band	**Pretendinous cord**	**MCPJ flexion contracture**
Spiral band + **lateral digital sheet** + **Grayson ligament** + pretendinous band	**Spiral cord**	**PIPJ contracture**; **displaces the neurovascular bundle centrally (midline), superficially and proximally** (**at risk during surgery**)
Lateral digital sheet	**Lateral cord**	**PIPJ / DIPJ contracture**
Natatory ligament	**Natatory cord**	**Web space contracture (adduction)**
Central cord (no normal precursor: from the pretendinous band continuation)	**Central cord**	**PIPJ contracture**

Normal structure	Diseased cord	Effect
Retrovascular band (Thomine)	**Retrovascular cord**	**DIPJ contracture** (residual after fasciectomy)
Abductor digiti minimi cord (little finger)	**Abductor digiti minimi cord**	**PIPJ contracture** of little finger
First web: proximal commissural ligament	**Commissural cords**	**First web contracture**

- **Cleland ligaments (dorsal to the neurovascular bundle) are NOT involved** in Dupuytren disease ("**Cleland's never Clings**"); **Grayson ligaments (volar to the bundle) ARE involved**.
- **Transverse fibres of the palmar aponeurosis: not involved** (usually).
- **Spiral cord: a spiral nerve should be suspected with a soft-tissue mass between the MCP and PIP flexion creases, a PIPJ contracture, and interdigital soft-tissue fullness.**
- **Fingers involved: ring (commonest) > little > middle > thumb > index**.

4. Clinical evaluation

- **Painless nodules in the palm (distal palmar crease, ring finger)**, **cords**, **skin pitting**, **flexion contractures (MCPJ, PIPJ)**.
- **Hueston tabletop test**: **inability to place the hand flat on a table = indication for surgery** (positive test).
- **Measure joint contractures with a goniometer.**
- **Tubiana classification (total flexion deformity per ray): stage 0** no lesion; **N** nodule without contracture; **stage 1: 0-45 degrees**, **2: 45-90**, **3: 90-135**, **4: over 135 degrees**.

5. Indications for intervention

- **MCPJ contracture of 30 degrees or more** (**MCPJ contractures correct well because the collateral ligaments are tight in flexion: no capsular contracture**).
- **Any PIPJ contracture (of about 15-20 degrees or more)** (**PIPJ contractures are harder to correct and recur: secondary capsular/volar plate/collateral ligament contracture, central slip attenuation**; **intervene earlier**).
- **Functional impairment, positive tabletop test.**
- **Nodules alone: not an indication** (unless painful).

6. Treatment options

Non-surgical

- **Splinting alone: ineffective**; **steroid injection** into painful nodules.
- **Radiotherapy** (**early nodular disease**: may slow progression).

Percutaneous needle fasciotomy (PNF/needle aponeurotomy)

- **Office-based**, local anaesthetic, divide the cord with a needle bevel at multiple points; **best for MCPJ contractures with well-defined palmar cords**.
- **Fast recovery, low complication rate**, **but recurrence higher (about 50-85% at 5 years)**; **repeatable**; **nerve injury and tendon injury risks** (keep the needle superficial; patient feedback).

Collagenase *Clostridium histolyticum* (CCH) injection

- **Injection into the cord (enzymatic lysis: type I and III collagen)**, **manipulation 24-72 hours later** to rupture the cord.
- **Results: good correction of MCPJ (about 75-80% to under 5 degrees), less for PIPJ**; **recurrence about 35-50% at 5 years**.
- **Complications: skin tears (commonest significant: during manipulation)**, **oedema, bruising, lymphadenopathy, pain**, **flexor**

tendon rupture (rare: ring/little finger PIPJ injections), **pulley rupture**, **CRPS**, **allergic reactions**.

- (Commercial availability has varied by country.)

Surgery

- **Limited (regional/selective) fasciectomy** (**the standard operation: excise the diseased fascia only**): **Bruner zig-zag incision or longitudinal incision with Z-plasties**; **recurrence about 20-40% at 5 years**; **PIPJ correction is less reliable**.
- **Segmental fasciectomy** (**excision of short segments of the cord through small incisions**).
- **Dermofasciectomy** (**excision of diseased fascia + overlying skin, replaced with a full-thickness skin graft**): **recurrent disease, diathesis, young patients with aggressive disease, skin involvement**; **lowest recurrence (under graft: "firebreak")**.
- **Open palm technique (McCash)**: palmar transverse incision left open to heal by secondary intention (**reduces haematoma, swelling and pain; useful in elderly**).
- **Additional PIPJ procedures** for residual contracture: **check-rein ligament release, volar plate release, accessory collateral ligament release**; **beware of central slip attenuation (residual extension lag)**.
- **Salvage: PIPJ arthrodesis**, **amputation** (multiple recurrences, severe little finger contractures), **dynamic external fixation (Digit Widget/TEC: continuous elongation: severe PIPJ contractures)**.
- **Post-operative: night extension splint (common, evidence of benefit limited)**, hand therapy.

Comparisons

- **Recurrence: dermofasciectomy < limited fasciectomy < collagenase ≈ needle fasciotomy.**

- **Recovery and complications: needle/collagenase faster but higher recurrence**; **surgery more durable but longer recovery**. **DISC trial (UK)**: collagenase vs limited fasciectomy: **surgery slightly better** clinically.

7. Complications of surgery

- **Digital nerve injury (about 2-3% primary; up to about 10% in revision)** (spiral cord), **digital artery injury (devascularisation: white finger after release: release tension (flex the finger), warm, topical vasodilators; microvascular repair)**, **haematoma**, **skin necrosis (flap tips)**, **infection**, **"flare reaction" (pain, swelling, stiffness: pseudo-CRPS: women)**, **CRPS (about 5%)**, **recurrence (true recurrence in the operated field) vs extension (new disease outside the field)**, **loss of flexion**, **PIPJ extension lag**.

Part B: Rheumatoid hand and wrist

8. General principles

- **Chronic symmetrical inflammatory polyarthritis**; **synovitis**: **pannus (proliferating synovium)** erodes cartilage, bone, **ligaments and tendons**; **tendon rupture by attrition or direct infiltration**.
- **Modern DMARDs/biologics** have **reduced the need for surgery**; **surgery: relieve pain, prevent tendon rupture (synovectomy), restore function, correct deformity**.
- **Priority of surgery (general principles): lower limb before upper limb**, **proximal before distal** (shoulder/elbow before wrist/hand), **wrist before fingers (stable wrist = foundation)**, **painful procedures first**; **joints before tendons** (or tendon transfers after joint stabilisation); **prophylactic tenosynovectomy** to prevent rupture (**over 4-6 months of persistent tenosynovitis despite medical therapy**).

- **Cervical spine assessment (atlantoaxial instability) before anaesthesia (flexion-extension radiographs)**; **perioperative DMARD management (methotrexate continued; biologics withheld around surgery)**.

9. Wrist

- **Caput ulnae syndrome**: **dorsal subluxation of the ulnar head** (**DRUJ synovitis with destruction of the TFCC and ECU subsheath; ECU subluxates volarly**: becomes a wrist flexor), **supination of the carpus** off the ulna, **radial deviation of the wrist** (**contributes to ulnar deviation of the fingers: "zig-zag deformity"**), **"piano key sign"** (ulnar head ballottement).
- **Carpal supination, volar subluxation, ulnar translocation of the carpus** (radiocarpal ligament destruction).
- **Vaughan-Jackson syndrome** (**attrition rupture of the ulnar extensors (EDM first, then EDC little and ring) over the prominent dorsal ulnar head**).
- **Mannerfelt-Norman syndrome** (**attrition rupture of the FPL** (then FDP index) **over a scaphoid/STT osteophyte** in the carpal tunnel).
- **Differentials of loss of finger extension in RA:**
  - **Extensor tendon rupture** (**tenodesis test: passive wrist flexion fails to extend the MCPJs**),
  - **PIN palsy** (**elbow synovitis compressing the PIN**; **tenodesis effect intact**),
  - **Extensor tendon subluxation into the ulnar gutter at the MCPJ** (**sagittal band attenuation: patient can maintain extension once passively extended**),
  - **MCPJ volar dislocation/subluxation** (cannot be passively extended).
- **Surgical options:**

- **Dorsal tenosynovectomy + extensor retinaculum transposed beneath the tendons** (protects them).
- **Distal ulna procedures: Darrach (distal ulna excision)** (**low-demand elderly; risk of ulnar stump instability/convergence**), **Sauvé-Kapandji (DRUJ arthrodesis + proximal ulnar pseudarthrosis)** (**younger; preserves ulnar support for the carpus: prevents ulnar translocation**), **ulnar head replacement**.
- **Tendon transfer/reconstruction for ruptures: single EDC rupture: side-to-side transfer to adjacent EDC**; **two tendons (little and ring): side-to-side + EIP transfer**; **three or more: FDS transfer (ring/middle) or free grafts**; **EPL rupture: EIP to EPL**; **FPL rupture: FDS ring transfer or IPJ fusion**.
- **Radiolunate (Chamay) fusion** (**prevents ulnar translocation; partial fusion when the midcarpal joint is preserved**).
- **Total wrist arthrodesis** (**gold standard for the destroyed rheumatoid wrist**; **Mannerfelt pin, Steinmann pin or plate**; **fuse in neutral or slight extension; if bilateral, consider one arthroplasty or fuse one in slight flexion for perineal hygiene**).
- **Total wrist arthroplasty**: **low-demand, bilateral disease, good bone stock**; preserves motion; higher complication rates.
- **Larsen grading** of radiological destruction (0-5).

10. MCP joints

- **Ulnar drift and volar subluxation** (**the commonest finger deformity in RA**):
  - **Synovitis stretches the radial sagittal band and collateral ligaments**,
  - **Extensor tendons sublux ulnarly into the valleys between the metacarpal heads**,

  - **Radial deviation of the wrist (with metacarpals) + ulnar pull of flexor tendons (enter the finger obliquely) + intrinsic contracture (ulnar intrinsics)** + **gravity/pinch forces** (**thumb pushes the index ulnarly**),
  - **Volar plate and accessory collateral ligament destruction**: **volar subluxation of the proximal phalanx**.
- **Treatment:**
  - **Early (no joint destruction): synovectomy, extensor tendon centralisation (sagittal band reefing), intrinsic release (cross-intrinsic transfer: ulnar intrinsic of one finger to the radial lateral band of the adjacent finger)**.
  - **Joint destruction: silicone MCPJ arthroplasty (Swanson: a flexible spacer; encapsulation)** with **soft-tissue rebalancing (radial collateral ligament reconstruction, centralisation, intrinsic release)**: **pain relief, improved arc and appearance (arc about 40-60 degrees, shifted into extension)**; **implant fracture (about 10-30% at 10 years), silicone synovitis (rare at the MCPJ), recurrent ulnar drift**. **Correct wrist radial deviation first** (otherwise recurrence).
  - **Pyrocarbon/surface replacement**: requires intact soft tissues (less used in RA).

11. Finger deformities

Swan-neck deformity (PIPJ hyperextension + DIPJ flexion)

- **Origins: MCPJ (volar subluxation/intrinsic tightness: commonest in RA)**, **PIPJ (volar plate laxity, synovitis, FDS rupture)**, **DIPJ (mallet: terminal tendon rupture)**.
- **Nalebuff classification (RA swan-neck):**
  - **Type I: PIPJ flexible in all MCPJ positions**: **splint (figure-of-eight/ring splint)**, **DIPJ fusion (mallet)**, **dermodesis, flexor**

tenodesis (FDS sling), SORL (spiral oblique retinacular ligament reconstruction),

- **Type II: PIPJ flexion limited with the MCPJ extended (intrinsic tightness: positive Bunnell-Littler/Finochietto test)**: **intrinsic release** + type I procedures,
- **Type III: PIPJ stiff in all positions, joint surface preserved**: **PIPJ manipulation, lateral band mobilisation, skin release (dorsal)**, **flexor tenosynovectomy**,
- **Type IV: PIPJ stiff with joint destruction**: **PIPJ arthrodesis (index/middle: lateral pinch stability) or arthroplasty (ring/little)**.

Boutonniere deformity (PIPJ flexion + DIPJ hyperextension)

- **PIPJ synovitis**: **central slip attenuation**, **lateral bands migrate volarly** (triangular ligament stretched).
- **Nalebuff/Millender stages:**
  - **Stage I (mild: 10-15 degree PIPJ extension lag, passively correctable)**: **splinting**, **synovectomy**, **Fowler/Dolphin terminal tendon tenotomy** (DIPJ hyperextension),
  - **Stage II (moderate: about 30-40 degree lag, passively correctable)**: **central slip reconstruction (Matev/Curtis), lateral band relocation**,
  - **Stage III (fixed, joint destroyed)**: **PIPJ arthrodesis (index/middle) or arthroplasty**.

Thumb deformities (Nalebuff classification: types I-VI)

- **Type I: boutonniere thumb (commonest: about 50-70%)**: **MCPJ flexion + IPJ hyperextension** (**MCPJ synovitis stretches the EPB insertion and dorsal capsule; EPL subluxes ulnarly/volarly**): **early: synovectomy + EPL rerouting (to the dorsal base of the proximal phalanx)**; **late: MCPJ arthrodesis (± IPJ arthrodesis/arthroplasty)**.
- **Type II: rare (boutonniere + CMCJ subluxation)**.

- **Type III: swan-neck thumb (second commonest)**: **CMCJ subluxation/adduction contracture**: **MCPJ hyperextension + IPJ flexion** (like OA): **CMCJ arthroplasty (trapeziectomy) + MCPJ fusion/volar capsulodesis**, **first web release**.
- **Type IV: gamekeeper's (UCL laxity: MCPJ radial deviation, secondary adduction)**: **UCL reconstruction or MCPJ arthrodesis**.
- **Type V: MCPJ volar plate laxity (MCPJ hyperextension, IPJ flexion; CMCJ normal)**: **volar capsulodesis/MCPJ fusion**.
- **Type VI: arthritis mutilans (opera-glass hand)**: **bone resorption, shortening, "telescoping"**: **arthrodesis with bone grafting** to restore length.

12. Other inflammatory hand conditions

- **Psoriatic arthritis: DIPJ involvement**, **pencil-in-cup deformity**, **arthritis mutilans**, **dactylitis (sausage digit)**, **nail pitting**, **asymmetric**; **stiffness (ankylosis) rather than instability**; **avoid surgery through active psoriatic plaques (infection risk)**; **arthrodesis in functional position** commonly.
- **Systemic lupus erythematosus (SLE): ligamentous laxity with reducible deformities (Jaccoud arthropathy: ulnar drift, swan neck) WITHOUT joint destruction**: **soft-tissue procedures fail (recurrence): arthrodesis often preferred**.
- **Scleroderma (systemic sclerosis): PIPJ flexion contractures** (**dorsal skin ulceration over the PIPJ**), **Raynaud phenomenon, digital ischaemic ulcers**, **calcinosis (CREST)**, **acro-osteolysis**; **PIPJ arthrodesis (with shortening)**, **digital sympathectomy (periarterial)**, **botulinum toxin for Raynaud**, **calcinosis excision**.
- **Gout: tophi (urate deposits)**, **flexor tenosynovitis, carpal tunnel syndrome**, **punched-out erosions with overhanging edges ("rat bite")**; **debulking of tophi** for skin ulceration, infection, nerve compression.

- **Osteoarthritis of the hand (comparison): Heberden nodes (DIPJ), Bouchard nodes (PIPJ), thumb CMCJ (basal joint) OA (Eaton-Littler stages I-IV: trapeziectomy ± LRTI for stages II-IV; CMCJ arthrodesis in young manual workers), mucous cyst (DIPJ OA ganglion: excise with osteophyte).**

Exam pearl: **Dupuytren: myofibroblast, type III collagen up, TGF-beta. Diathesis: early onset, family history, bilateral, ectopic (Garrod, Ledderhose, Peyronie). Cleland ligament spared; Grayson involved. Spiral cord (pretendinous band + spiral band + lateral digital sheet + Grayson) displaces the NV bundle central, superficial, proximal.**

Exam pearl: **Operate for MCPJ over 30 degrees, any PIPJ contracture, positive tabletop test. Recurrence: dermofasciectomy lowest; needle and collagenase highest. Collagenase: skin tears; tendon rupture rare. White finger after release: flex finger, warm.**

Exam pearl: **RA: caput ulnae (dorsal ulnar head, ECU volar), Vaughan-Jackson (EDM first), Mannerfelt (FPL over scaphoid). Loss of finger extension: rupture (tenodesis test negative), PIN palsy (tenodesis intact), subluxed tendon (holds once placed), MCPJ dislocation. Darrach (elderly) vs Sauvé-Kapandji (young). Wrist fusion is gold standard; correct wrist radial deviation before MCPJ silicone arthroplasty.**

Exam pearl: **Swan-neck Nalebuff I flexible, II intrinsic tight, III stiff/joint preserved, IV destroyed. Boutonniere thumb (type I) commonest RA thumb deformity. SLE: reducible deformities without destruction (Jaccoud); soft-tissue procedures fail. Psoriatic: DIPJ, pencil-in-cup. Scleroderma: PIPJ flexion, calcinosis.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.